



Accessibility as a Service

Shared accessibility infrastructure for public institutions — an API, a widget & an extension

Read aloud · Translate · Speak-to-fill · Dyslexia-friendly — for every citizen

AaaS

SMART ODISHA HACKATHON '25

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Theme: Accessibility · E-commerce & Cyber Security

Software category · Odiapreneur track

One API · One widget · One extension — accessibility for every citizen in Odia (ଓଡ଼ିଆ), Hindi & English

THE CHALLENGE

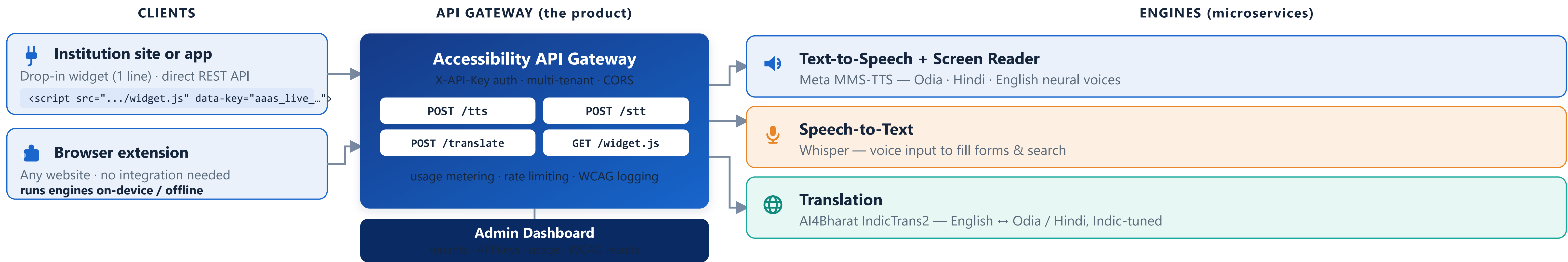
- Govt & education sites are largely unusable for blind, low-vision & dyslexic users — screen readers struggle.
- Content is English-first; Odia-only citizens are shut out.
- Every institution rebuilding accessibility from scratch is slow, costly and inconsistent.

OUR SOLUTION — one shared platform

- Institutions integrate ONCE — a secure REST API + a drop-in widget (a single <script> line).
- A browser extension brings it to ANY site — no integration.
- Neural Odia/Hindi/English voice; on-device & offline-ready; WCAG-aligned, DPDP-conscious, India-first.

PLATFORM ARCHITECTURE

One way in → one Accessibility API Gateway → shared engines (also run on-device, offline)



On-device option: the extension runs MMS-TTS · Whisper · IndicTrans2 in-browser (offline, works on strict-CSP sites). **Built on the platform:** Accessible Examination module — extra time, audio support, dyslexia-friendly fonts.

KEY FEATURES

- Read this page**
Neural Odia / Hindi / English TTS
- Translate this page**
IndicTrans2, rewritten in place
- Speak to fill**
Voice → form fields (Whisper)
- Dyslexia mode**
Conjunct-safe Odia spacing
- Hover to speak**
Point at anything to hear it
- Works everywhere**
Any website, via the extension

ONE PLATFORM, SIX PARTS

- 1 Speech-to-Text
 - 2 Text-to-Speech + Screen Reader
 - 3 Accessibility API Gateway
 - 4 Translation — IndicTrans2
 - 5 Accessible Examination module
 - 6 Admin Dashboard
- TECH** MV3 Extension FastAPI Gateway Meta MMS-TTS Whisper AI4Bharat IndicTrans2 ONNX / WebAssembly

WCAG 2.1 AA aligned

Multi-tenant API service

DPDP-conscious data flow

Offline-capable

Odisha-first

For: students with disabilities, PwDs & non-English-speaking citizens
Built for: schools, universities, government offices & examination boards